Vibe Coding Educativo

Responsible vibe coding

A guide to publishing educational materials created with vibe coding

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Introduction

What educational vibe coding is

Vibe coding, a name proposed by Andrej Karpathy in February 2025, consists of creating computer programs by explaining to an artificial intelligence (AI) model what one wants to obtain, without having to write the code.

EDUCATIONAL VIBE CODING

Educational vibe coding, as understood in this guide, is **the creation of resources and applications for teaching and learning by talking in natural language with an AI system, which writes the code, while the person defines the educational purpose and checks the result**.

Why this guide

In the Telegram group Vibe Coding Educativo, more and more materials created with AI are being shared, sometimes without a licence, sending students' data to external services or without anyone else being able to make use of the code. The person who publishes them is accountable for them, even if the code was written by the AI, and the guide proposes, based on the Open Knowledge Decalogue, a responsible way of doing it.

RESPONSIBLE EDUCATIONAL VIBE CODING

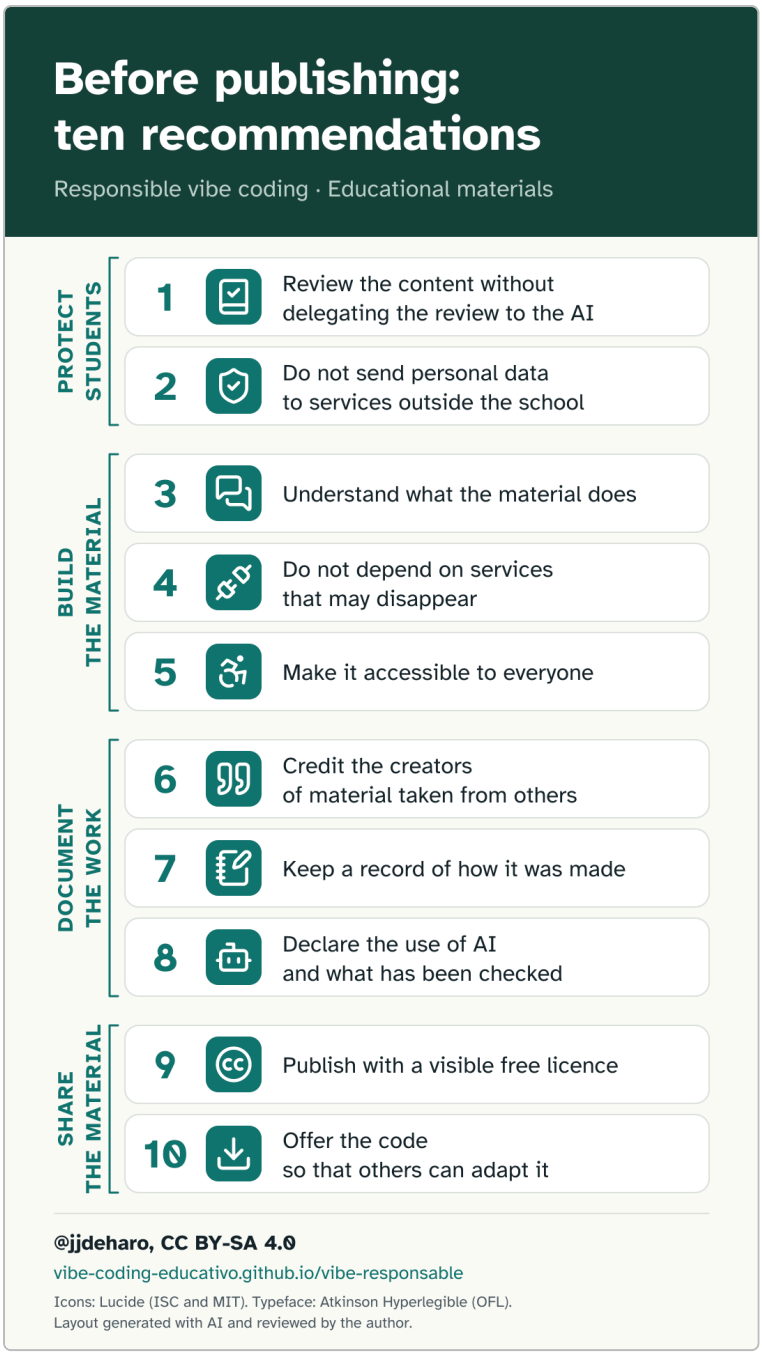
Responsible educational vibe coding (VCER, from its Spanish initials) is the kind that **publishes materials that are safe for students, accessible, durable, transparent about how they were made and free for other people to reuse**.

How to start

1 When creating a resource.
Give the AI the instructions file, which tells it how to follow the recommendations.

2 When it is finished.
Ask it for the VCER evaluation with the evaluation file, which points out what is missing.

3 To publish.
With these two steps, the AI takes care of the minimum of each recommendation, except for reviewing the content, which is the person's job.



Watch the animation

Before publishing: ten recommendations

PROTECT STUDENTS

1 Review the content without delegating the review to the AI

Artificial intelligence (AI) can make mistakes with complete naturalness, and an error in a simulation or a quiz ends up as mistaken learning. Before publishing, the material has to be used as students would use it, checking the concepts, the data and the answers it accepts as correct. This review cannot be delegated, since responsibility for what is taught lies with the person who publishes it.

Minimum. Go through the material from beginning to end, also with wrong answers, and use subject expertise to check every result. After each significant change, repeat the walkthrough with a checklist that the AI itself can draft, since a small modification can break something that already worked.

Recommended. Ask the AI to turn that checklist into automated tests and run them after every change.

2 Do not send personal data to services outside the school

Students' names, marks, voices or images are personal data. In Spain, the Spanish Data Protection Agency states in its guide for schools (https://www.aepd.es/documento/guia-centros-educativos.pdf) that teachers must use the tools provided by the school or the education authority, and that content a teacher publishes on their own, outside the school, is their own responsibility. Other countries have different rules, but the precaution is the same. The simplest way to comply is for the material not to ask for data. When a tool needs to identify students, such as a gradebook, the data must stay on the teacher's device or in the systems the school has decided to use. Particular care is needed with platforms that easily add user accounts and databases, since the data are then stored on third-party servers.

Minimum. Do not ask for real names or anything that identifies a person, unless the tool needs it to do its job, and in that case store it only on the device. Ask the AI whether the application sends information to any server. If the material opens inside a platform, check whether it requires registration or a minimum age before sending the link to students, since they are being taken to a third party's service.

Recommended. Publish the material on a site that students can open without registering, and check that the code contains no web addresses of unrecognised services. When the program is going to manage students' data in the school's systems, the decision belongs to the school, and a technical review is advisable before putting it into use.

3 Understand what the material does

If nobody understands how a resource works, it cannot be fixed when it fails, and in practice it stops being open. Meeting this point does not require knowing how to program, since it is enough to be able to describe briefly what the application does, what it stores and whether it communicates with any external service.

Minimum. Ask the AI to explain in plain language what the application does and whether it stores or sends anything, and check that the explanation matches what can be observed when using it. Also ask for the material's own code to be commented and readable, since code compressed into endless lines is reason enough not to publish. Well-known libraries that are included are the exception, as they are usually distributed that way.

Recommended. Add a document explaining how the project is organised and what each file is for.

4 Do not depend on services that may disappear

A resource that embeds content from another website, or loads parts from third-party servers, stops working when those services change or close. The same happens with the platform where the material was created, since the shared link lasts as long as the company decides.

Minimum. Keep a copy of the material's code on the teacher's own computer, and update it when it changes.

Recommended. Ask the AI to load whatever the material needs from outside from well-known services and to record it in the decision log, and keep the project's own images, texts and data inside it.

5 Make it accessible to everyone

Materials generated with AI tend towards the showy, and decorative effects are often an obstacle for some students. An accessible resource can be used without a mouse, can be understood without relying on colour and reads well on a small screen.

Minimum. Ask the AI from the start to follow the accessibility guidelines, and test the result with the keyboard only, with enlarged text and on a phone.

Recommended. Ask the AI to check accessibility with an automated tool and fix what it detects. Coding agents can do this unaided, since they install the tool, run it and apply the fixes.

DOCUMENT THE WORK

6 Credit the creators of material taken from others

The images, texts, sounds and pieces of software that are incorporated have authors and licences, even if the AI placed them there. The attribution must go inside the material itself, so that it travels with it when it circulates out of context.

Minimum. State the author, source and licence of each third-party element, and replace those that do not allow reuse.

Recommended. Also check the licences of the software libraries included, since some of them determine the licence of the whole.

7 Keep a record of how it was made

With AI, work moves very fast, and a few weeks later nobody remembers why each decision was taken. Keeping that record makes it possible to resume the work, to explain it to someone else and not to undo by mistake something that had a reason.

Minimum. Keep a document with the important decisions and their reasons. The AI itself writes it when asked at the end of each working session, and all that remains is to review and save it.

Recommended. Ask the AI to keep a decision log or ADR (*Architecture Decision Record*) inside the project. The AI writes it from what is decided in the conversation, with the context, the discarded alternatives and the consequences of each decision and, for technical ones, what it is based on and how it was checked. All that remains is to check that what is recorded matches what was decided.

8 Declare the use of AI and what has been checked

In a resource created with vibe coding, the code is the work of the AI, and usually nobody has reviewed it line by line. The people who reuse it need to know this in order to decide how far they can trust it. It is therefore advisable to state which tool was used to create it and, above all, what the person who publishes it has checked, such as the correctness of the content, how it works or how data are handled.

Minimum. One or two sentences inside the material, next to the licence, with the tool used and what has been checked.

Recommended. The same statement in the project documentation, with a link to the decision log from point 7, which is what explains how the material was made.

SHARE THE MATERIAL

9 Publish with a visible free licence

Works are protected by copyright automatically, so a resource without a licence cannot be safely reused even if it has been published. A free licence tells other people that they may use it, adapt it and share it, and on what conditions. Code and content need different licences, and what the AI generates raises questions of authorship that are dealt with in its chapter.

Minimum. Write the author and the licence inside the material itself, in a visible place, for example in the footer.

Recommended. Add the licence file to the project, with a free software licence for the code, such as AGPL v3 or MIT, and a free Creative Commons (CC) licence, such as CC BY-SA or CC BY, for the content.

10 Offer the code so that others can adapt it

The licence gives permission, but the material must also be obtainable in a form that allows work to be done on it. That way someone else can adapt it to their classroom without asking anyone for anything.

Minimum. Offer the code for copying or downloading, together with the decision log from point 7, so that someone else can continue it.

Recommended. Publish it in an open repository, with an explanation of how to use it and how to modify it, and ask the AI to mark each published version with a version tag, which makes it possible to retrieve the exact code of that version later.

1 Review the content without delegating the review to the AI

What needs to be done

Minimum. Go through the material from beginning to end, also with wrong answers, and use subject expertise to check every result. After each significant change, repeat the walkthrough with a checklist that the AI itself can draft, since a small modification can break something that already worked.

Recommended. Ask the AI to turn that checklist into automated tests and run them after every change.

How artificial intelligence makes mistakes

The texts generated by artificial intelligence (AI) sound good, the explanations are convincing and the references look real. However, they are sometimes false. The risk does not lie in the obvious error, which anyone detects, but in the plausible error: the date that is almost right, the quotation that almost existed, the figure that almost matches. Martín Núñez Calleja, from the National Centre for Curriculum Development in Non-Proprietary Systems (CEDEC), sums it up clearly in his talk «Crear REA con eXeLearning en tiempos de IA» (https://descargas.intef.es/cedec/formacion/SL_REA_IA_julio26/html/fiabilidad-del-contenido.html) (Creating OER with eXeLearning in times of AI), devoted to open educational resources (OER): **«In the classroom, an error that nobody detects is not an error: it is mistaken learning»**.

In an interactive resource the problem multiplies, because the error is not read once but repeated with every student who uses it. A simulator with a misapplied formula, a quiz that accepts a wrong answer as correct or an explanation that reverses a cause-and-effect relationship teach that error as many times as the material is opened.

What is reviewed in the material

It is advisable to review the whole material before publishing it, and not only the part that was expressly requested. The check covers four aspects:

- **The content.** The concepts, data, dates, units and formulas that appear in the explanations.
- **The activities.** The questions, the answer options and, above all, which ones are accepted as correct. A badly designed distractor can be as harmful as a wrong answer.
- **The feedback.** What the material replies when students get it wrong, which can only be seen by trying incorrect answers.
- **Behaviour in edge cases.** What happens when an answer is left blank, a negative number is typed, a huge value is entered or a button is pressed twice in a row.

The way to do this is to go through the material as students would, also with wrong answers, and not just look at the initial screen. Many faults only appear on reaching the end of the activity or on repeating it.

Reviewing after each change

With AI it is easy to modify a resource that has already been reviewed, and that ease carries a risk. A change that seems small, such as adding a question or changing the design, can break another part that

already worked, even if it apparently has nothing to do with what was requested. These faults are known as regressions, and the initial review does not detect them, since it was carried out on the previous version.

To detect them, **the review becomes a checklist that is repeated after each significant change**. The list includes the main walkthroughs and the edge cases that have already been checked, and the AI itself can draft it from what has been tested. For a quiz, for example, it could be the following:

- The quiz can be completed from beginning to end.
- A wrong answer shows the intended feedback.
- A blank answer does not cause errors.
- On restarting, the activity returns to its initial state.
- It works with the keyboard only.

When the project allows it, the AI can turn those checks into automated tests and run them after each change. Coding agents do this unaided. These tests only confirm that the material still works as before, and the correctness of what it teaches still depends on the review described above. The AI can take care of repeating the checks, although deciding which behaviours must be kept is up to the person who knows the material.

Pedagogical responsibility

AI speeds up production, but **pedagogical responsibility still lies with the person who publishes the material**. This is also stated in the «Guía sobre el uso de la inteligencia artificial en el ámbito educativo» (https://code.intef.es/wp-content/uploads/2026/09/ACTUALIZACI%C3%93N-GU%C3%8DA-DE-LA-IA-DEF-1-SEPT-2026-Publicable-v5.pdf) (Guide on the use of artificial intelligence in education) of Spain's National Institute of Educational Technologies and Teacher Training (INTEF), in its version 2.0 of September 2026, which includes human oversight and responsibility among its ethical principles: teachers must keep control over the use of AI, and educational decisions cannot depend on automated systems.

This has a practical consequence for the rest of the guide. The AI can take care of almost all the recommendations, from the licence to accessibility, but not this one. It can point out what should be checked, and even warn about what it is unsure of, but it cannot certify that what the material teaches is correct. That is why, in the VCER evaluation, the AI scores only the errors it detects and points out what the person must review.

Reviewing after publication

The review does not end on publication. The people who use the material find faults that its author has not seen, because they test it with other students, on other devices and with other questions. It is therefore advisable to make it known in a teaching community and to pay attention to the comments of the people who try it. In the Telegram group Vibe Coding Educativo (https://t.me/vceduca), the «¡Comparte tu App!» (Share your app) section is meant for presenting published programs, and the comments they receive help to correct and improve them.

The educational value of the material

Accurate data are the minimum, not the goal. A resource can contain no errors at all and still be poor from a teaching point of view. The article «Mantener la "A" de abierto en los REA en tiempos de IA» (http

s://cedec.intef.es/mantener-la-a-de-abierto-en-los-rea-en-tiempos-de-ia/) (Keeping the “O” of open in OER in times of AI) points out three aspects that no code replaces.

The first is the clarity of the learning objective, so that students know at all times what they are expected to learn and why. The second is formative feedback, which does not consist only of saying whether an answer is correct, but of explaining why, offering a hint and inviting another attempt. The third is methodological coherence, that is, each activity responding to a teaching decision and not to what the AI proposed by default.

The question worth asking when reviewing is not whether the material has turned out showy, but whether it helps students learn better. That judgement belongs to the person who knows the subject and to the students, and it is their contribution to educational vibe coding.

2 Do not send personal data to services outside the school

What needs to be done

Minimum. Do not ask for real names or anything that identifies a person, unless the tool needs it to do its job, and in that case store it only on the device. Ask the AI whether the application sends information to any server. If the material opens inside a platform, check whether it requires registration or a minimum age before sending the link to students, since they are being taken to a third party's service.

Recommended. Publish the material on a site that students can open without registering, and check that the code contains no web addresses of unrecognised services. When the program is going to manage students' data in the school's systems, the decision belongs to the school, and a technical review is advisable before putting it into use.

What needs to be done depends on the type of material. The table summarises the four cases explained in this chapter, and each one links to its section.

If the material...	Then...
is intended for students or for open publication	it needs no data, and should not ask for any
is a teacher's tool with identified students	the data stay on the teacher's device, in line with the school's rules
collects students' answers	the results reach the teacher without going through services outside the school
is provided by the school to its teachers	the data go to the school's systems, and the decision belongs to the school

Personal data in an educational resource

Personal data are any information that makes it possible to identify a person. In an educational resource they include students' names, email addresses, marks, voices or images, and also the answers to an activity when they are stored linked to a name. A quiz that asks for the name and sends it to a server together with the answers is processing personal data.

Most students are minors, and data protection rules are stricter in that case. In the European Union, the General Data Protection Regulation (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32016R0679) sets at 16 the age at which a person can consent to the processing of their own data in online services, and allows each country to lower it to 13. In Spain, Organic Law 3/2018 (https://www.boe.es/buscar/act.php?id=BOE-A-2018-16673) sets it at 14, and below that age consent is given by the families. That age only counts when the processing is based on consent. According to the guide for schools (https://www.aepd.es/documento/guia-centros-educativos.pdf) of the Spanish Data Protection Agency, schools do not, as a general rule, need

consent to process students' data in the exercise of their educational function, although they must provide information about that processing.

Even when consent is not needed, or students can give it themselves, good practice advises that families receive accurate information about what their children do and the tools they use. That information should state which materials are used, whether they collect any data and for what purpose, so that the family knows the material before it is used in the classroom. The Orientaciones sobre el uso de herramientas digitales en el ámbito educativo desde la perspectiva de la protección de datos (https://intef.es/wp-content/uploads/2026/07/Orientaciones_sobre_el_uso_de_herramientas_digital.pdf) (Guidelines on the use of digital tools in education from a data protection perspective), from Spain's National Institute of Educational Technologies and Teacher Training (INTEF), consider it good practice, although not compulsory, and ask for that information to reach families through the school's channels, and not through social networks or instant messaging.

The decision about students' data

Publishing a resource that sends students' data to a server **is not a decision each teacher can take on their own**. In Spain, the same guide states that education authorities and schools must have instructions for teachers' use of technologies, and that teachers must use those provided by the authority or the school. In a state school, the controller of students' data is normally the education authority. By contrast, for what a teacher publishes outside their teaching role at the school, the controller is the teacher.

The consequence is that a resource created on a teacher's own initiative must not store students' data in a service that the school has not provided. INTEF's guidelines go in the same direction. As a general rule, teachers should not use tools other than those provided by the authority or the school, and any other tool that processes personal data requires a prior evaluation, which the school consults with its data protection office, and the permission of the authority or the school. The document includes an evaluation model that can also be applied to one's own material. Other countries have different rules, but the precaution is the same: before collecting students' data, the school must be involved.

Materials that need no data

The simplest way to comply is for the material not to collect data. The European Regulation establishes the principle of data minimisation, according to which data must be limited to what is necessary for their purpose, and requires data protection by design. The Guidance on AI and children (https://www.unicef.org/innocenti/reports/policy-guidance-ai-children) of the United Nations Children's Fund (UNICEF), which has a worldwide scope, recommends the same: minimising data collection and adopting a privacy-by-design approach.

In a resource created with vibe coding, this translates into a few decisions that can be requested from the AI from the start:

- **No identification.** The material does not ask for the name. If several people need to be told apart, an alias that does not identify them is enough.
- **Everything in the browser.** Answers and progress are stored on the device itself and are not sent to any server.
- **No user accounts.** The material is used without registering with any service.

- **No tracking.** The material includes no analytics or tracking tools. At most, a count of visits per page that does not tell readers apart, kept by the publisher and explained on the website, like the one the Spanish Data Protection Agency (AEPD) (https://www.aepd.es/guias/guia-cookies-analiticas-externas.pdf) exempts from requiring consent (sections II and III.A).

With a resource built this way, the person who publishes it receives no data from students, and the data do not leave the device on which it is used. If a teacher stores their students' marks in it, they are still processing personal data, but they do so within their educational function and without any outside service being involved.

Tools that need to identify students

Some materials need to identify students to do their job, such as a gradebook, a seating plan or a group generator. These tools can be created and published, since monitoring students is part of the educational function. **The condition is that the data remain under the control of the teacher and their school.**

The Spanish Data Protection Agency's guide accepts that teachers use applications on their personal devices, provided they respect the privacy policy defined by the school or the education authority. It considers it especially important that such use does not involve transferring students' data to the service provider, for the provider to use for its own purposes or to store permanently.

In a tool created with vibe coding, this condition is met when the data are stored only on the teacher's device, either in the browser or in a file that is downloaded and loaded again. The published application contains no data, because each teacher enters their own and they do not leave their computer, so the author of the tool processes nobody's data. It is also advisable for the tool to allow the information to be shared or printed without the names, and for each teacher to use it in line with their school's rules, just as they would a paper notebook or a spreadsheet.

One example is the Cuaderno del Profesorado (https://github.com/imanlost/CuadernoProfesorado-v1.0) (Teacher's Gradebook), by Imanol Lostalé, a complete teacher's gradebook that works without a server: the database is stored in the browser or in a file on the teacher's disk, with its backups.

Programs that collect students' answers

Another case is the program a teacher creates to check their students' knowledge, detect misconceptions or follow their progress. The initiative is individual, but the data are generated by students on their own devices and have to reach the teacher. Assessment is part of the educational function, so the program is legitimate, and what needs care is the path the results follow. INTEF's guidelines reserve data with legal effects, such as the mark for a test, for the school's tools, so the mark that counts is recorded on the school's platform. There are several ways to solve this without sending data to services outside the school, which can be combined:

- **The result is handed in through the school's channels.** The program shows the result at the end, or saves it in a file, and students hand it in through the school's platform, like any other assignment. This is what OpenWorksheets (https://openworksheets.github.io/) does, a free application for interactive worksheets with no server or accounts, in which students download a submission file when they finish.

- **Students identify themselves with a code.** The program does not ask for the name, but for a code that only the teacher can link to a particular student. The European Regulation calls this technique pseudonymisation, and requires that correspondence to be kept separately. INTEF's guidelines give this same example of pseudonymised use through a code, on condition that it does not allow students to be identified again. In the Plantilla correctora digital (PCD) (https://jjdeharo.github.io/pcd/) (Digital answer-sheet marker), for multiple-choice exams, the only identification field accepts a code instead of the name.
- **The results are collected in a school spreadsheet.** When automatic collection is needed, each teacher deploys their own copy of the program in the account the school provides them with, and not in a personal account, so that the results reach a spreadsheet that only they control. The program's author publishes a template and receives no data. The school should be aware of it. An example is the Quiz del Sistema Solar (https://poblogguizar.github.io/quiz/) (Solar System Quiz), by Pablo G. Guízar, designed for each teacher to deploy with their own spreadsheet, which keeps the correct answers out of the browser. The same author explains how to do it safely in his security guide for applications with Google Sheets (https://poblogguizar.github.io/apps-with-google-sheets/).
- **The submission travels encrypted.** The program encrypts the result with a key belonging to the teacher, so that students can encrypt but not decrypt, and only the teacher reads it with their password. Even if the file is sent through an insecure channel, its content is unreadable. OpenWorksheets (https://openworksheets.github.io/) offers this encryption as an option, and also warns if a submission file has been tampered with.
- **The mark is managed by the school's platform.** The program is exported in a standard format and uploaded to the platform the school already uses, which is the one that records the results. One of these formats is SCORM (*Sharable Content Object Reference Model*). This is what the Generador SCORM de Certificado de Finalización (https://github.com/PabloGGuizar/generador-scorm-de-certificado-de-finalizacion) (SCORM completion certificate generator) does, by Pablo G. Guízar, which obtains the student's name from the platform itself and records there that the course has been completed.
- **Answers travel directly between devices.** In live activities, such as a classroom quiz game, students' devices can connect to the teacher's without the answers being stored on any server. An intermediary service usually helps to establish the connection, and it sees technical identifiers but not the content. This is how the Buzzer WebRTC (https://poblogguizar.github.io/buzzer-webrtc/) works, by Pablo G. Guízar, a buzzer for quick-response activities. WebRTC (*Web Real-Time Communication*) is the browser technology that allows that direct connection.

These programs are deterministic, that is, they always apply the same rules to the answers, and the analysis is done on the device itself. The situation changes if the program sends the answers to an AI service to assess them. In that case the data reach a third party and, in addition, the European AI Regulation (https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng) classifies AI systems intended to evaluate learning outcomes as high-risk, which entails significant obligations for their provider and for the school that uses them. Following the amendment approved in July 2026 through Regulation (EU) 2026/1744 (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32026R1744), those obligations will apply from 2 December 2027.

Programs provided by the school to teachers

A school or an education authority may decide that a program should manage students' data in its own systems, for example some shared spreadsheets with marks. In that case the data leave the browser, and that is legitimate, because the decision is taken by the controller of those data. The Spanish Data Protection Agency's guide states that schools must know the applications they are going to use, their

privacy policy and their terms of use before using them. When an external provider is involved, it acts as a processor and may only process the data in accordance with the school's instructions.

A program created with vibe coding can fulfil this function with three precautions. The first is that the data are stored on the platform the school already uses, without adding new services. The second is that the decision goes through the school management and the data protection officer, and not through an individual teacher. The third is security: the code of these programs was written by the AI and usually nobody has reviewed it, so a program that handles real data deserves a review by a person with technical knowledge before it is put into use.

When the program is published as a template for other schools to use, it is advisable to accompany it with a notice stating who is responsible for the data. One model is that of IAGuar (https://elprofedelabata.es/iaguar/), an application for managing teachers' cover duties, whose author states that he has no access to the data of copies deployed by third parties and that the controller is the school that sets up its own copy.

Sending data to other servers

Another risk lies in code that sends information without its author knowing. The article «Mantener la "A" de abierto en los REA en tiempos de IA» (https://cedec.intef.es/mantener-la-a-de-abierto-en-los-rea-en-tiempos-de-ia/) (Keeping the "O" of open in OER in times of AI), which the National Centre for Curriculum Development in Non-Proprietary Systems (CEDEC) devotes to open educational resources (OER), describes an illustrative case: a mathematics quiz that, on completion, sent the answers to an external domain. The teacher thought it was a usage counter, and in fact that domain was collecting minors' data without the consent required by the Regulation. The same article warns about materials that ask for permission to use the camera, the microphone or the location without a clear educational purpose.

App-building platforms deserve special attention, since they easily add user accounts and databases, and the data are then stored on a company's servers. The same happens when the material is opened within a chatbot's website: students enter a third party's service, which may require registration or a minimum age. If they are used, INTEF's guidelines recommend that the servers be in the European Union and that the provider clearly state who the data controller is, for what purpose it uses the data, to whom it discloses them and for how long it keeps them.

To check a resource it is not necessary to read the code. A first check consists of asking the AI to list all the external addresses that appear in it and to explain what each one is for, as the VCER evaluation asks. **An address that is not recognised is reason enough not to publish until it has been clarified**. If the material has user accounts or connects to a server, it is also advisable to ask the AI to review how those data are accessed and how they are protected.

3 Understand what the material does

What needs to be done

Minimum. Ask the AI to explain in plain language what the application does and whether it stores or sends anything, and check that the explanation matches what can be observed when using it. Also ask for the material's own code to be commented and readable, since code compressed into endless lines is reason enough not to publish. Well-known libraries that are included are the exception, as they are usually distributed that way.

Recommended. Add a document explaining how the project is organised and what each file is for.

Understanding the code

An educational resource is open when someone else can download it, understand it, modify it and improve it. The article «Mantener la "A" de abierto en los REA en tiempos de IA» (https://cedec.intef.es/mantener-la-a-de-abierto-en-los-rea-en-tiempos-de-ia/) (Keeping the "O" of open in OER in times of AI), which the National Centre for Curriculum Development in Non-Proprietary Systems (CEDEC) devotes to open educational resources (OER), warns that a resource with hundreds of lines of code that nobody understands, not even the person who inserted them, has stopped being open in essence, even if its licence says otherwise. In his talk «Crear REA con eXeLearning en tiempos de IA» (https://descargas.intef.es/cedec/formacion/SL_REA_IA_julio26/html/la-ia-y-el-codigo.html) (Creating OER with eXeLearning in times of AI), Martín Núñez Calleja puts it this way: «We have the source code. But not the understanding».

The problem is a practical one, since a resource that works today may stop working after a browser update, and if nobody understands how it is made, it cannot be fixed either. The same article proposes a simple rule: **if what the code does cannot be explained in two sentences, the resource is not yet ready to be published**.

A brief description of the material

Meeting this recommendation does not require knowing how to program, since it is enough to be able to say three things about the material in ordinary words: **what it does, what it stores and whether it communicates with any external service**. An explanation of this kind would be the following: «The simulator calculates the acceleration of a body on an inclined plane from the chosen angle and material, and draws the forces. It stores no data and does not connect to any service».

The way to obtain it is to ask the AI itself, in plain language, and then check that it matches what can be observed when using the material. If the artificial intelligence (AI) states that nothing is stored and the material remembers the previous day's answers, the explanation is not correct and it must be clarified before publishing. The VCER evaluation includes this request in its point 3.

Four checks without knowing how to program

The CEDEC article proposes four questions to detect problematic code without needing to understand it

line by line:

- **Whether it can be read.** Legitimate code has recognisable words, spaces and lines of reasonable length. A line of hundreds of characters without spaces, with letters and numbers mixed together, indicates that the code is obfuscated.
- **Whether it refers to external addresses.** It is advisable to look for the web addresses that appear in the code and investigate those that are not recognised.
- **Whether it asks for permissions.** Access to the camera, the microphone, the location or the clipboard is only acceptable when the material justifies it with a clear educational purpose.
- **Whether its author can explain it.** This is the two-sentence rule.

All four can be entrusted to the AI, which can locate the addresses and permissions in the code. The same article summarises these criteria in a traffic light that classifies code as safe, moderately risky and dangerous.

Readable, commented code

Code generated by AI without comments works like a black box. When it includes comments in natural language, other people can understand, modify and maintain it more easily. It is advisable to ask for this from the start, with the comments in the language of the material and with understandable variable names. One example is the inclined plane with friction simulator (https://onio72.github.io/iesmajuelo/bach/fq1/planoincroz/), by Antonio González García, whose code is in a single file, divided into labelled sections and commented in Spanish.

Code compressed into endless lines is reason enough not to publish, with the exception of well-known libraries, which are usually distributed that way. When the material is a project with several files, it is advisable to add a document explaining how it is organised and what each file is for. This is what Pablo G. Guízar does in the repository of his Solar System Quiz (https://github.com/PabloGGuizar/quiz), which describes the program's architecture, the path of the data and the function of each file.

4 Do not depend on services that may disappear

What needs to be done

Minimum. Keep a copy of the material's code on the teacher's own computer, and update it when it changes.

Recommended. Ask the AI to load whatever the material needs from outside from well-known services and to record it in the decision log, and keep the project's own images, texts and data inside it.

A resource's dependencies

A resource depends on an external service when it needs something that is not inside it in order to work. The most common forms are content embedded from another website, programming libraries and typefaces loaded from third-party servers, and connections with online services. The platform where the material was created is also a dependency, when the material only exists within it.

These dependencies are not visible when using the material. They are discovered by reading the code or by asking the artificial intelligence (AI) to list them, which is what point 4 of the VCER evaluation asks for.

Changes in external services

An external service may change its terms, become paid or close, and the material that depended on it **stops working without its author having touched anything**. The article «Mantener la "A" de abierto en los REA en tiempos de IA» (https://cedec.intef.es/mantener-la-a-de-abierto-en-los-rea-en-tiempos-de-ia/) (Keeping the "O" of open in OER in times of AI), which the National Centre for Curriculum Development in Non-Proprietary Systems (CEDEC) devotes to open educational resources (OER), illustrates this with the case of a resource that embeds an interactive map from an external platform. If the platform withdraws free presentations, the resource shows a blank box, and the teacher cannot recover the content because they do not have the original file. The article recommends reserving embedded content for videos and similar cases.

The risk is not only that the service disappears. In 2024, the polyfill.io domain, from which more than one hundred thousand websites loaded a widely used library, changed hands and began serving malicious code without the affected pages having changed anything, as documented by the security company Sansec (https://sansec.io/research/polyfill-supply-chain-attack). The same CEDEC article warns of another risk specific to AI-generated code: invented libraries, or libraries impersonated by others with an almost identical name.

A platform's link

When the material has been created on a chatbot's website or on an app-building platform, the shared link lasts as long as the company decides. A change in the service, in its terms or in the teacher's account can render that link useless, and with it all the pages that have embedded it.

The minimum is to keep a copy of the material's code on one's own computer and update it when it changes. With that copy the material can be recovered, published elsewhere or further developed with

another tool. For it to be useful, the material has to be a page that opens on its own in the browser. The most widely used chatbots, such as ChatGPT, Gemini or Claude, often generate the application as a component built with React, a widely used programming library. That component only works within the chatbot's own website. That is why the instructions file for the AI asks them for an HTML page.

What is loaded from outside

Programming from scratch what a well-known library already solves is not realistic, and neither is hosting inside the material everything it uses. Libraries that display formulas, charts or maps, and typefaces, can be loaded from outside, provided they come from a well-known service and are recorded. A well-known service does not guarantee that it will stay the same either, and that is why it is advisable to check what stops working when the material is opened offline.

The recommended step is to ask the AI to load these resources from well-known services and to record them in the decision log of recommendation 7, with their licence. That list is not meant for the teacher, but for fixing or adapting the material later on, a task that will often be done by an AI again. What cannot be retrieved from elsewhere, such as one's own images, texts and data, should be inside the material, or at least kept in a copy, and not only embedded from another platform.

What matters most to the teacher is a practical consequence: **whether the downloaded resource will keep working when opened on a computer without an internet connection**, for example in a classroom without a connection. The check requires no technical knowledge, since it consists of opening the material, disconnecting the device from the network and loading it again. One example is Tantrix (https://felipsarroca.github.io/jocs/Tantrix/), a game by Felip Sarroca that can be installed as an application and works offline after the first visit. The VCER evaluation asks the AI to explain it in simple words, of the kind «if you open it without internet, the formulas will not display».

5 Make it accessible to everyone

What needs to be done

Minimum. Ask the AI from the start to follow the accessibility guidelines, and test the result with the keyboard only, with enlarged text and on a phone.

Recommended. Ask the AI to check accessibility with an automated tool and fix what it detects. Coding agents can do this unaided, since they install the tool, run it and apply the fixes.

Accessibility in an educational resource

A resource is accessible when people with disabilities can perceive it, understand it, navigate it and interact with it. The Web Accessibility Initiative (https://www.w3.org/WAI/fundamentals/accessibility-intro/) of the *World Wide Web Consortium* (W3C), the consortium that develops web standards, points out that it covers visual, auditory, physical, cognitive, neurological and speech disabilities, and that it also benefits other people, such as those using a mobile phone, those with a slow connection or those with a temporary limitation.

In education it is not an optional improvement, since a resource that some students cannot use leaves those people out of the activity. The Recommendation on Open Educational Resources (OER) (https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer) of the United Nations Educational, Scientific and Cultural Organization (UNESCO) places effective, equitable and inclusive access among its objectives, and in the European Union accessibility is a legal requirement for public sector websites, established by Directive 2016/2102 (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32016L2102).

Decorative effects

Artificial intelligence (AI) makes it effortless to add entrance animations, custom cursors or particle backgrounds. The article «Mantener la “A” de abierto en los REA en tiempos de IA» (https://cedec.intef.es/mantener-la-a-de-abierto-en-los-rea-en-tiempos-de-ia/) (Keeping the “O” of open in OER in times of AI), from the National Centre for Curriculum Development in Non-Proprietary Systems (CEDEC), points out that these effects add no educational value, and that they increase the file size, slow down loading on slow connections and can interfere with screen readers and other assistive technologies.

It is advisable to do without what merely decorates, and to ask the AI to respect the device’s reduced motion preference when an animation does have a function.

Tests within reach of any teacher

There are four tests that require no tools or technical knowledge:

- **With the keyboard only.** Go through the material with the Tab key and activate the controls with the Enter key. All controls must be reachable, in a logical order, and it must be visible which one has the focus.
- **With enlarged text.** Zoom the page to double size from the browser. The content must still be readable without texts being cut off or overlapping.

- **On a mobile phone.** The material must fit the width of the screen, and the buttons must be pressable with a finger.
- **Without relying on colour.** A right or wrong answer must not be distinguishable only by colour. It must be accompanied by text or a symbol.

Review with the help of AI

Much of this work can be entrusted to the AI. The minimum is to ask it from the start for the material to follow the Web Content Accessibility Guidelines (https://www.w3.org/WAI/standards-guidelines/wcag/), the international standard, which is organised into four principles: content must be perceivable, operable, understandable and robust. This is what the instructions file for the AI does.

The recommended step is also to ask it to check the result with an automated tool and fix what it detects. Coding agents can do this unaided, since they install the tool, run it and apply the fixes. The W3C warns, however, that no tool alone can determine whether a site meets accessibility standards (https://www.w3.org/WAI/test-evaluate/), so **manual tests are still necessary**.

One example is the interactive guide «MCD y mcm» (https://felipsarroca.github.io/edu-apps/MCD-mcm/), by Felip Sarroca, devoted to the greatest common divisor and the least common multiple, in Catalan. It passes the automated check without a single warning, has all its controls within keyboard reach, with visible focus, states its language and adapts to a mobile phone screen.

6 Credit the creators of material taken from others

What needs to be done

Minimum. State the author, source and licence of each third-party element, and replace those that do not allow reuse.

Recommended. Also check the licences of the software libraries included, since some of them determine the licence of the whole.

Third-party material within one's own material

A resource created with artificial intelligence (AI) is rarely entirely new. It usually incorporates images, icons, sounds, typefaces, texts or programming libraries that have their own authors. The AI adds them naturally when asked for a result, and it does not always say where they come from or the conditions under which they can be used.

Recognising authorship is a matter of respect for other people's work, and also a right. In Spain, the Intellectual Property Law (https://www.boe.es/buscar/act.php?id=BOE-A-1996-8930), in its article 14, includes among the author's moral rights that of demanding recognition as the author of the work. Free licences maintain that condition. The Creative Commons Attribution-ShareAlike (https://creativecommons.org/licenses/by-sa/4.0/deed.en) licence, abbreviated CC BY-SA, for example allows the work to be copied and adapted as long as appropriate credit is given, a link to the licence is provided and any changes are indicated.

Information to include in an attribution

Creative Commons recommends in its recommended practices for attribution (https://wiki.creativecommons.org/wiki/Recommended_practices_for_attribution) that each third-party element be accompanied by four pieces of information:

- **The title** of the work, if it has one.
- **The author**, with the name or username of its creator.
- **The source**, with the link to the place it was taken from.
- **The licence**, with its name and the link to its text.

When the element has been modified, it is advisable to say so. A complete attribution would take this form: «Title of the image, by Author's name, taken from Wikimedia Commons, licensed under CC BY-SA 4.0», with links to the original image and to the licence.

Where the attribution goes

The attribution must go inside the material itself, and not on the page that links to it or in a separate message. An educational resource is downloaded, embedded in another page or uploaded to a school's platform, and if the attribution does not travel with it, it is lost at the first step. The minimum is a credits section in the footer or on a screen of its own, with the four pieces of information for each element. This guide does so on its credits and licences page, which lists the icons and the typeface it uses, with their authors and licences.

What the AI adds on its own

It is advisable to ask the AI which third-party elements it has incorporated and under which licence, since it does not always say. This is what point 6 of the VCER evaluation asks for. There are three cases that deserve attention:

- **Images linked from another website.** The AI may place an image by means of its address on another site, without its licence being stated. It is advisable to replace it with one whose licence is known and keep it inside the material.
- **AI-generated images.** They have no author to credit, but it is advisable to state that they are generated and with which tool, as recommendation 8 asks.
- **Programming libraries.** Each one has its licence, and some determine the licence of the whole. Copyleft licences, such as the GPL (*General Public License*), require the program that incorporates them to be distributed under the same licence. Others, such as MIT, named after the Massachusetts Institute of Technology, ask for the authorship notice to be kept.

When an element does not allow reuse, or its licence cannot be found out, **the prudent thing is to replace it**. The AI can propose alternatives with a free licence, and it is advisable to check at the original source that the licence is what it says.

7 Keep a record of how it was made

What needs to be done

Minimum. Keep a document with the important decisions and their reasons. The AI itself writes it when asked at the end of each working session, and all that remains is to review and save it.

Recommended. Ask the AI to keep a decision log or ADR (*Architecture Decision Record*) inside the project. The AI writes it from what is decided in the conversation, with the context, the discarded alternatives and the consequences of each decision and, for technical ones, what it is based on and how it was checked. All that remains is to check that what is recorded matches what was decided.

The reason for each decision

With artificial intelligence (AI), programming moves very fast, and **a few weeks later nobody remembers why each decision was taken**. Ernesto Serrano, from the eXeLearning team, describes it in his talk «Inteligencia artificial: programar, documentar y no acabar en un berenjenal» (https://erseco.github.io/talks/charlas/2026-07-06-selia-ia-programar-documentar/unit/index.html) (Artificial intelligence: programming, documenting and not ending up in a mess): the code is there, but the why is not, and nobody remembers which alternatives were discarded or on what grounds. AI does not cause that disorder, although it makes it arrive sooner.

Keeping the record serves three purposes. It makes it possible to resume the work after some time without having to reconstruct it, to explain it to someone else who wants to continue it, and to avoid undoing in good faith a decision that had a reason. In an open educational resource it has a fourth use, since it shows the people who reuse it how it was made, which is what completes the statement of recommendation 8.

The decision log

The usual way to keep it in software development is the architecture decision record, or ADR (*Architecture Decision Record*). Each important decision is recorded in a short document that includes four things:

- **The context.** The situation that makes a decision necessary.
- **The decision.** What is done, in enough detail to apply it.
- **The discarded alternatives.** Each one with the reason it was not chosen, which is the part that avoids repeating the debate later on.
- **The consequences.** What improves and what gets worse.

A decision that stops being valid is not deleted, but marked as superseded by the new one, so that the record is preserved. In the project presented in the talk, each record also notes with which AI tool and which model the decision was taken, which makes it possible to change tools without losing the history.

In that same project, each record also includes the evidence on which the decision is based, the known risks and the way it was validated. The talk sums it up in the rule «no source, no claim»: every fact refers

to the official documentation, to a specific version of the code or to a test that anyone can repeat. This precaution is especially useful with AI, since it can write a convincing justification for a decision that starts from a false premise. The recorded evidence lets the person check it without redoing the work. Whatever could not be verified is recorded as a hypothesis pending validation, so that nobody later takes it for a checked fact.

The work of the AI and the work of the person

The log is not an additional task for the teacher, since the AI writes it from what is decided in the conversation, and the person checks that what is recorded matches what was decided. The talk sums it up by saying that the AI makes proposals and the person makes the decisions.

Nor is the log reconstructed at the end, since a resource usually comes out of many working sessions spread over different days, and putting those conversations back together afterwards is unfeasible. **The log is written at the moment the decision is taken**, and that is why it is preserved from one session to the next. Coding agents are told once in their instructions file, and they keep it up in all sessions. It is advisable to ask for it by name, for example «keep a decision log with ADRs», since the AI knows the format and applies it without further explanation. On a chatbot's website, the minimum is to ask the AI at the end of each session to record that day's decisions in a document that is kept and added to.

An example from the community

The interactive guide «Elige tu IA» (https://explikarlos.github.io/elige-ia/) (Choose your AI), published as expliCarlos, has a decision log (https://github.com/explikarlos/elige-ia/blob/main/docs/decisions/ADR-001-static-pages.md) in its repository. The first record explains why the application is static and has no server. Among the discarded alternatives is a database, which would have allowed answers to be synchronised, but which was discarded because it increased privacy risks, cost and maintenance. Anyone who takes up that project knows that the absence of a server is the result of a considered decision.

This guide also has its own log (https://github.com/Vibe-Coding-Educativo/vibe-responsable/tree/main/docs/adr), which records the decisions about its sources, its website and its examples.

8 Declare the use of AI and what has been checked

What needs to be done

Minimum. One or two sentences inside the material, next to the licence, with the tool used and what has been checked.

Recommended. The same statement in the project documentation, with a link to the decision log from point 7, which is what explains how the material was made.

What the statement is for

In a resource created with vibe coding the code is written by the artificial intelligence (AI), and usually nobody has reviewed it line by line. The people who reuse it need to know this in order to decide how far they can trust it, just as they need to know which part its author has checked. The statement on the use of AI answers those two questions in a couple of sentences.

Transparency is one of the ethical principles of the «Guía sobre el uso de la inteligencia artificial en el ámbito educativo» (https://code.intef.es/wp-content/uploads/2026/09/ACTUALIZACI%C3%93N-GU%C3%8DA-DE-LA-IA-DEF-1-SEPT-2026-Publicable-v5.pdf) (Guide on the use of artificial intelligence in education) of Spain's National Institute of Educational Technologies and Teacher Training (INTEF), according to which students and teachers must know when AI is used, how it works and what its limitations are. The National Centre for Curriculum Development in Non-Proprietary Systems (CEDEC), in its article on the Conference on Free Software and Open Artificial Intelligence (seLIA) (https://cedec.intef.es/cedec-en-selia-dos-miradas-sobre-ia-software-libre-y-recursos-educativos-abiertos/), proposes sharing with each resource which tools were used, what was reviewed by hand and what was generated automatically. It sums it up by saying that this documentation «is not a layer of bureaucracy, it is a layer of honesty» towards the community that is going to reuse the resource.

The legal obligation to declare

In the European Union, the AI Regulation (https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng) establishes in its article 50 some transparency obligations that apply from 2 August 2026. Most of them fall on the providers of AI systems, who must mark generated content so that its origin can be detected. Providers whose systems were already on the market before that date have until 2 December 2026 to do so, according to the amendment introduced by Regulation (EU) 2026/1744 (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32026R1744). For the people who use those systems, it requires disclosure in two cases: when they disseminate generated or manipulated images, audio or video that resemble real people, objects, places, entities or events and could be taken as authentic, what the Regulation calls a deep fake, and when they publish texts to inform about matters of public interest without these having gone through human review or editorial control for which someone is responsible.

An educational resource reviewed by its author does not usually fall into those cases, unless it includes realistic images, audio or video of that kind, such as a recreation of a historical event that could be taken as authentic. Outside that case, **declaring the use of AI is good practice and not a legal requirement**. Other countries have different rules, and it is advisable to consult them.

The content of the statement

The statement must be brief and specific. It should indicate three things:

- **The tool used.** The name of the chatbot, the platform or the agent with which the material was created.
- **What the person has checked.** The correctness of the content, how it works, how data are handled or accessibility. This is the most informative part, because it varies from one resource to another.
- **The date.** Tools change quickly, and the date makes it possible to place the material in time.

A statement of this kind could be «Created with [name of the tool] in September 2026. The content has been reviewed and the operation tested. It collects no data». **The minimum is for it to appear inside the material itself, next to the licence**, so that it accompanies it when it is downloaded or embedded in another page. José Luis Miralles Bono does this in applications such as ScoreSpacer (https://jlmirallesb.github.io/ScoreSpacer/), whose footer states that it was created with the help of a specific AI and that all processing takes place in the browser. When the material is a project with several files, the same statement goes in its documentation, with a link to the decision log of recommendation 7, which is what explains how it was made.

AI involvement in teaching content

In code, the involvement of the AI is always the same, so assigning a level to it adds nothing. In teaching content the opposite happens. The questions of a quiz may have been written entirely by the teacher, or the AI may have generated them and the teacher merely validated them, and that difference matters to the person who is going to use them in their classroom.

To indicate it, the Framework for integrating generative AI into educational tasks (MIAE) (https://jjdeharo.github.io/miae/) can be used, which describes AI involvement at six levels, from the completely human work of level 0 to human supervision of an autonomous AI at level 5. To do so it is enough to add the level to the statement, for example «Content at MIAE level 2, reviewed by the author». This guide applies it to itself, and its credits page states that it was produced at level 4, that of advanced collaboration between the person and the AI.

9 Publish with a visible free licence

What needs to be done

Minimum. Write the author and the licence inside the material itself, in a visible place, for example in the footer.

Recommended. Add the licence file to the project, with a free software licence for the code, such as AGPL v3 or MIT, and a free Creative Commons (CC) licence, such as CC BY-SA or CC BY, for the content.

Materials without a licence

Works are protected by copyright from the moment they are created, without needing to register them or to say so. The consequence is that **a resource published without a licence has all rights reserved**, so other people can view it, but cannot copy, adapt or share it safely, even if its author has no objection.

The licence is what turns a published resource into a shared resource. The Recommendation on Open Educational Resources (OER) (https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer) of the United Nations Educational, Scientific and Cultural Organization (UNESCO), adopted in 2019, defines them as materials that reside in the public domain or have been released under an open licence that permits no-cost access, re-use, re-purposing, adaptation and redistribution by others. Therefore, a resource without an open licence is not an open educational resource, even if it can be freely consulted.

Licences for content and for code

A resource created with vibe coding brings together two types of work, which use different licences. The content is the texts, questions, images and explanations, and is usually published under a Creative Commons (CC) licence. The code is what makes the material work, and it uses licences designed for computer programs.

Not all Creative Commons licences are free, in the sense set out in the Definition of Free Cultural Works (https://freedomdefined.org/Definition): a free work can be studied, copied, modified and redistributed by anyone and for any purpose. Creative Commons itself uses this definition (https://creativecommons.org/public-domain/freeworks/) to distinguish its free licences, CC BY and CC BY-SA, from those that are not. Each condition is indicated by two letters. BY requires authorship to be credited, and SA (*ShareAlike*) requires the same licence to be kept in adaptations. Those that include the NC (*NonCommercial*) condition, which prohibits commercial uses, or the ND (*NoDerivatives*) condition, which prohibits adaptations, limit precisely what one wants to allow. For content, therefore, two free licences remain:

- **CC BY (https://creativecommons.org/licenses/by/4.0/deed.en).** It allows any use, including adapting the material, with the sole condition of crediting the author. It gives the widest dissemination, although improved versions may be published under a more restrictive licence.
- **CC BY-SA (https://creativecommons.org/licenses/by-sa/4.0/deed.en).** It adds the obligation to publish adaptations under the same licence, so that improvements remain free for the whole educational community.

It is advisable to avoid the NC condition, even though it may seem the most prudent, because it is not always clear what a commercial use is. Creative Commons itself states (https://creativecommons.org/faq/#does-my-use-violate-the-noncommercial-clause-of-the-licenses) that it depends on the circumstances and the intention with which the material is used, and not on the type of organisation, and that it cannot clarify which uses are commercial. The use by a private school or by a platform with advertising, for example, is debatable.

If the aim is that nobody appropriates the material, that protection is given by the SA condition. A company may use a resource under CC BY-SA, but it has to publish its adaptations under the same licence, so that the improvements return to the community. The NC condition, on the other hand, closes two doors. The material is left out of repositories that only accept free resources (https://commons.wikimedia.org/wiki/Commons:Licensing), and it cannot be mixed with other free materials. According to the Creative Commons FAQ (https://creativecommons.org/faq/#if-i-derive-or-adapt-material-offered-under-a-creative-commons-license-which-cc-license-s-can-i-use), an adaptation of SA material has to be published under the same licence, so a text or an image under CC BY-SA cannot be incorporated into a resource under CC BY-NC-SA.

For code, the most widely used free software licences are four:

- **MIT (https://opensource.org/license/mit).** The simplest, which allows any use provided the authorship and licence notice is kept.
- **Apache 2.0 (https://www.apache.org/licenses/LICENSE-2.0).** Equally permissive, but it asks for changes to be indicated and includes a patent grant.
- **MPL 2.0 (https://www.mozilla.org/MPL/2.0/)** (*Mozilla Public License*). It requires modifications to files under this licence to be shared, even if they are combined with closed code.
- **AGPL v3 (https://www.gnu.org/licenses/agpl-3.0.html)** (*Affero General Public License*). It requires the code of modified versions to be shared under the same licence, also when they are offered as a service on a web page and not distributed. That last condition takes effect when the program runs on a server, since a resource that runs entirely in the browser hands its code to the person who opens it.

This guide recommends **CC BY-SA for content and AGPL v3 for code**, because it ensures that improvements return to the educational community. For maximum adoption, CC BY with MIT or Apache 2.0 can be chosen. The licence generator (https://jjdeharo.github.io/licencias-libres/) prepares the notice for any of these combinations, as text or as HTML ready to paste. The article «Licencias libres para docentes» (https://educacion.bilateria.org/licencias-libres-para-docentes-apps-educativas-y-materiales-didacticos/) (Free licences for teachers, in Spanish) goes deeper into the subject, with comparative tables, practical cases and answers to the most common questions.

Displaying the licence

The licence must be visible to fulfil its function. The minimum is for the material itself to state, in a visible place such as the footer, the authorship, the name of the licence and the link to its official text. That way the licence accompanies the material when it is downloaded or embedded in another page.

When the material is a project with several files, it is also advisable to add the file with the full text of the code licence and to mention both licences in the documentation. In a repository, the `LICENSE` file contains the licence of the code and another one, for example `LICENSE-CONTENIDOS`, that of the content. At the top of each code file a line such as `SPDX-License-Identifier: AGPL-3.0-or-later` is also advisable, which identifies the licence in a standardised way. All this can be requested from the artificial

intelligence (AI) from the start, as the instructions file for the AI does, so that the material is generated with its licence already in place.

Authorship of what the AI generates

The question of who owns what the AI generates has two parts. The first is whether the company offering the AI claims any rights, and the answer is no. The terms of the main services, consulted in September 2026, agree on this point. OpenAI (https://openai.com/policies/eu-terms-of-use/) states that the user owns the output and assigns to them all the rights it might have over it. Anthropic (https://www.anthropic.com/legal/consumer-terms) likewise assigns to the user its rights over the outputs. Google (https://policies.google.com/terms?hl=en) states that it will not claim ownership of original content generated with its services. The material, therefore, can be published, shared and licensed without the company having any objection.

The second part is whether there is a copyright over what is generated that can be enforced against other people, and that is not decided by the terms of use, but by the law of each country. The companies themselves assign their rights «if any», because nobody can assign what they do not have. In Spain, the Intellectual Property Law (https://www.boe.es/buscar/act.php?id=BOE-A-1996-8930) considers the author to be the natural person who creates a work, and the question is whether the person giving instructions to the AI creates the work.

There are two positions on that question. The first holds that AI is a tool, like the brush or the camera, and that authorship belongs to the person who wields it. It is the one taken by British law (https://www.legislation.gov.uk/ukpga/1988/48/section/9) since 1988, which attributes the authorship of a computer-generated work to the person who makes the arrangements necessary for its creation, and the one followed in 2023 by the Beijing Internet Court (https://legalblogs.wolterskluwer.com/copyright-blog/beijing-internet-court-grants-copyright-to-ai-generated-image-for-the-first-time/) when it recognised copyright in an image generated with AI, because of the decisions the user took in choosing and adjusting the instructions. The second position holds that authorship requires the person's creative decisions to be perceptible in the result. It is that of the United States Copyright Office (https://www.copyright.gov/newsnet/2025/1060.html), which concluded in 2025 that protection does not extend to the mere formulation of instructions, but does extend to what the person has created and can be recognised in the result, and to the creative arrangements and modifications they make to it. In Spain there is still no ruling, but the first European courts that have ruled follow the second position. In 2026, the Higher Regional Court of Düsseldorf (https://nrwe.justiz.nrw.de/olgs/duesseldorf/j2026/20_W_2_26_Urteil_20260402.html), in line with two other German courts, held that an image generated with AI is only a work when it reflects specific creative decisions of the person, and that general instructions, however many, or choosing among several proposals, are not enough. The Court of Justice of the European Union has not yet ruled, so the question is not settled.

This guide does not need to resolve it, because its recommendation is the same under either position: **publish the material with a free licence and state that it was created with AI**, as recommendation 8 asks. Under the first position, the licence covers the whole material. Under the second, it covers the person's contribution, and the rest would be left without rights that prevent its reuse. In both cases the material can be shared. It should be borne in mind that, under the second position, conditions such as share-alike could only be enforced on the human contribution. In educational vibe coding that contribution is extensive, since the person decides what is built, provides the content and corrects the result over many sessions, and recommendations 7 and 8 serve to record it.

10 Offer the code so that others can adapt it

What needs to be done

Minimum. Offer the code for copying or downloading, together with the decision log from point 7, so that someone else can continue it.

Recommended. Publish it in an open repository, with an explanation of how to use it and how to modify it, and ask the AI to mark each published version with a version tag, which makes it possible to retrieve the exact code of that version later.

Openness in practice

A free licence gives permission to reuse a resource, but **the permission falls short if the material cannot be obtained in a form that allows work to be done on it**. The Recommendation on Open Educational Resources (OER) (https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer) of the United Nations Educational, Scientific and Cultural Organization (UNESCO) does not speak only of access, but of re-use, re-purposing, adaptation and redistribution. Free software puts it in a similar way with its four freedoms (https://www.gnu.org/philosophy/free-sw.en.html), which include the freedom to study how the program works and change it, and the freedom to distribute copies of modified versions, for which access to the code is a necessary condition. The Declaration on libre knowledge (https://wikieducator.org/Declaration_on_libre_knowledge) extends the same idea to any knowledge resource: it is free when anyone can use it for any purpose, learn from it, copy it, adapt it and share the result for the benefit of the community.

The article «Mantener la "A" de abierto en los REA en tiempos de IA» (https://cedec.intef.es/mantener-la-a-de-abierto-en-los-rea-en-tiempos-de-ia/) (Keeping the "O" of open in OER in times of AI), from the National Centre for Curriculum Development in Non-Proprietary Systems (CEDEC), makes this concrete in a principle of adaptable simplicity, according to which it is better to prefer what is functional and simple to what is complex and closed, and the quality criterion of a resource is that it can be reused.

Obtaining the material

The minimum is for the material to offer its code for copying or downloading. On a chatbot's website and on app-building platforms the code can usually be viewed, but other people only receive a link, so it is advisable to add to the material itself the way of obtaining it, or to publish it separately. Together with the code, it is advisable to offer the decision log of recommendation 7, which allows someone else to continue it without having to guess why it is made that way.

The recommended step is to publish the project in an open repository, with an explanation of how to use it and how to modify it. A repository also allows other people to propose improvements and the author to incorporate them.

In a repository the code keeps changing after it is published, so the link to the project always leads to the most recent version. To retrieve later the exact version that was published, used in class or evaluated, **it is advisable to mark each published version with a version tag**, or *tag* in Git terminology, such as v1.0 or v1.1. The AI creates it when publishing, and it is enough to tell it once in its instructions. A tag that has already been published is not moved or reused, because it would stop pointing to what was pub-

lished. Next to it, the identifier of the *commit* can be recorded, the code that Git assigns to each saved state of the project, which serves as an exact reference even if the tag were changed by mistake.

A resource that is easy to adapt

A resource is more reusable when the content is separated from the functionality. A quiz whose questions are in a list at the beginning of the code, or in a separate file, can be adapted to another subject by changing that list, without touching the rest. It is advisable to ask the artificial intelligence (AI) for this from the start, together with the commented code of recommendation 3 and the absence of dependencies of recommendation 4, which are the other two conditions that make adaptation easier.

It also helps to state in the documentation which parts are meant to be changed, such as the texts, the colours or the language. Someone who wants to translate the material, or adjust it to another educational level, thus finds where to start.

Tools and levels

Four families of tools

Not everyone works with the same means, and the tools for creating programs with artificial intelligence (AI) are very varied. Broadly speaking, they can be grouped into four families, presented below with some examples of those available in 2026.

A chatbot's website.

General assistants such as ChatGPT (https://chatgpt.com/), Gemini (https://gemini.google.com/) or Claude (https://claude.ai/), which create the application within the conversation itself. The material is hosted on the platform and shared through a link.

App-building platforms.

Services such as Lovable (https://lovable.dev/), Bolt (https://bolt.new/), Replit (https://replit.com/), Google AI Studio (https://aistudio.google.com/) or Canva (https://www.canva.com/), which generate, host and publish the application from the browser.

Code editors with AI.

Programs such as Visual Studio Code (https://code.visualstudio.com/) or Cursor (https://cursor.com/), which are installed on the computer and include an assistant that writes and modifies the project files.

Coding agents.

Programs such as Codex CLI (https://developers.openai.com/codex/cli), Antigravity (agy) (https://antigravity.google/) or Claude Code (https://claude.com/product/claude-code), which receive instructions in natural language and work directly with the files on the computer.

The way of working

To follow this guide, what matters is the way of working that each tool allows. In the first two families the material lives on a company's servers, and the company decides how long it is kept. On a chatbot's website it is also a single piece. The code can usually be viewed, and even modified, but it cannot be accompanied by other files, such as the licence or the documentation, nor can changes be tracked conveniently. In the last two families the material is a folder of files on one's own computer, which can be published on one's own site or in a repository such as GitHub (https://github.com/), where every change is recorded. The article «Consolas de IA: qué son y cómo se instalan» (https://educacion.bilateria.org/consolas-de-ia-en-2026-que-son-como-se-instalan-y-que-cuestan-ahora/) (in Spanish) explains how to start working with coding agents.

The minimum and the recommended

For this reason, each recommendation is set out in two levels of guidance. **The minimum** is what can be done with any tool, even when the material has been created and is shared within a chatbot's website, and it is enough to publish responsibly. **The recommended** is what is also advisable, and it requires the material to be a project with its own files, published in a repository or on one's own site. App-building platforms also tend to work with multi-file projects, and some allow them to be sent to an external repository, so in that case both levels can be followed. When there is no difference between one way of working and another, a single level is given.

Instructions for creating with AI

How to use them

The guide's recommendations can be given directly to the artificial intelligence (AI), so that the work of following them does not fall on the person. There is a file of instructions for this, which can be downloaded and used with any tool. It contains only what the guide asks for; what the material must do is described separately, as always.

The file is handed over at the start:

- **On a chatbot's website**, it is attached to the first message, together with the description of the material to be created and the instruction «follow the instructions in the attached file».
- **In a coding agent or a code editor with AI**, it is saved in the project folder as an instructions file, and the AI takes it into account in every session.
- **On app-building platforms**, it is pasted into the project's permanent instructions, which many offer in their settings or as a file in the project itself.

If the AI has not been told who the author is or where the resource will be published, the instructions require it to ask before starting. When finished, it is advisable to ask it, in the same conversation, for the VCER evaluation, which points out what is missing. That review does not replace the person's own, since the model also makes mistakes when reviewing its own work.

To create a resource

This is the file given to the AI when starting to create a resource. It can be downloaded to attach it or to save it in the project folder, or its text can be copied and pasted at the beginning of the conversation.

```
# Instructions for creating an open educational resource

These instructions come from the guide «Responsible vibe coding», for
publishing educational materials created with vibe coding
(https://vibe-coding-educativo.github.io/vibe-responsable/en/).
Follow them throughout the work, together with what I ask you about the
material I want to create. The material will be published openly, so that
other people can use and adapt it.

## Before starting

If I have not told you, ask me:
- The authorship that should appear (a name or a username).
- Where it will be published: a chatbot's website, an app-building platform,
  or a repository or one's own site.

## How it must be built

- Licence: CC BY-SA 4.0 for the content and AGPL v3 for the code, unless I
  indicate others.
- If we are working on a chatbot's website, make the material as an HTML
  page that works on its own when opened in the browser, and not as a
  component that only works inside the chatbot.
- You may use libraries, typefaces and other external resources when they
  save work or improve the result. Load them from a well-known, stable
```

```
  service, and record them in the decision log, with their licence. If any
  part of the material would stop working when the downloaded copy is
  opened on a computer without an internet connection, tell me in simple
  words, for example: «if you open it without internet, the formulas will
  not display».
- Personal data: do not ask for the name or any data that identifies a
  person, unless the tool needs it to do its job, such as a gradebook. In
  that case store it only on the device and offer the option to export or
  print without the names. If the program collects students' answers, show
  the result at the end or let it be downloaded for handing in, and identify
  each person with a code instead of their name. Do not send anything to any
  server or add analytics. If I ask you for a visit counter, it should only
  add up the visits to each page on my own server, storing nothing that tells
  readers apart, neither their IP address nor where they came from, and the
  website should explain it.
- Accessible, following the Web Content Accessibility Guidelines (WCAG):
  usable with the keyboard only, with a logical tab order, labels on the
  controls, alternative text on images, sufficient contrast, not relying on
  colour to understand anything and readable on a mobile phone screen.
- Readable code, commented in the language of the material, not
  compressed, with understandable variable names.
- In the footer of the material: the title, the authorship, the licence with
  its link and a short statement on the use of AI, with the tool, the month
  and year, and what I have checked (ask me before publishing).
- If you incorporate images, sounds or code fragments by other people, use
  only material whose licence allows reuse, and state its authorship, source
  and licence within the material itself.
- If the program is going to store students' data in the school's systems,
  do not write keys or passwords in the code that reaches the browser, make
  sure only the teacher or the school can read what is collected, and tell
  me what a person with technical knowledge should review before it is put
  into use.
- After each change, check again what already worked: run the tests if there
  are any or, if you cannot, tell me which checks I should repeat.

## If it is published in a repository or on one's own site

- Add a LICENSE file with the licence of the code and another with that of
  the content, and state the licence at the top of each code file with an
  SPDX-License-Identifier line.
- Add a document explaining how the project is organised and what each file
  is for.
- Keep a decision log (ADR) inside the project, with one file per decision
  recording the context, the discarded alternatives and the consequences.
  Record in it every decision we take, without waiting for me to ask. For
  technical decisions, also record what you base them on (official
  documentation, a specific version of the code or a test that can be
  repeated), the known risks and how you checked it. Do not invent sources
  or tests: mark whatever you have not been able to check as a hypothesis
  pending validation.
- Check accessibility with an automated tool and fix what it detects.
- When we publish a version, mark it with a version tag (v1.0, v1.1…) and do
  not move or reuse a tag once it has been published.

## When you finish the material
```

```
- Briefly describe what the material does, what it stores and whether it
  communicates with any external service.
- Prepare a short checklist with the main walkthroughs and the edge cases,
  to repeat after each change. If the project allows it, turn it into
  automated tests.
- Write a short note with the important decisions you have taken and the
  reason for each one, or summarise them from the decision log if there is
  one.
- Tell me what I should check myself, starting with the correctness of the
  content.
```

The VCER evaluation

What the VCER evaluation is

VCER stands for «vibe coding educativo responsable», the Spanish for responsible educational vibe coding. It is the evaluation proposed by the guide «Responsible vibe coding» to check whether an educational resource meets the essentials of the guide before it is published or used. It is suitable for any finished resource, one's own or someone else's, whether or not it was created following the guide.

The evaluation is carried out by an AI using a file of instructions that is downloaded from the guide. The AI reviews the code of the resource, scores each point of the rubric with 2, 1 or 0, justifies each score and proposes the three improvements that would raise the result the most.

How the result is calculated

The percentage is the sum of the scores divided by the maximum possible for the points the AI has been able to check. Content and personal data are disqualifying. From the percentage and those two points, one of these three results is obtained:

Result	When it applies	What it means
Recommended	70 % or more, with content and personal data scored and with no 0 in either of them or in accessibility	The resource meets the essentials of the guide and can be used or published. The proposed improvements complete it.
Needs improvement	Less than 70 %, or when the content or the personal data could not be checked, or accessibility has a 0	The resource has flaws that should be fixed before publishing or recommending it, although none of them rules it out. The report says what they are and what remains to be checked.
Not recommended	A 0 in content or in personal data, whatever the percentage	The resource has obvious errors in what it teaches or sends students' data to services outside the school. It should not be used with students or published until it is fixed.

Limits of the evaluation

The AI only detects the errors it sees, especially in the content, and the result may vary depending on the model used. That is why the score is indicative and does not replace a review by a person. In addition, the result applies to the version of the resource that was evaluated, on the date of the evaluation.

How to evaluate a resource

The evaluation file, which appears below, is given to the AI together with the resource to be evaluated, and the AI is asked to «evaluate this resource according to the instructions». How to give it depends on where the resource is:

- **If it was created on a chatbot's website or on an app-building platform**, the file is attached, or its text pasted, in the same conversation, once the material is finished.
- **If it is in a folder on the computer or in a repository**, the file is given to a coding agent or a code editor with AI opened in that folder. This is the most complete way, because the AI can read all the files of the project and, if it can run code, test the resource in the browser.
- **If only the published website is available**, one's own or someone else's, a conversation with the AI is opened and the evaluation file and the code of the page, saved from the browser, are attached. If the website is made up of several files, the AI will only see the ones it is given, so it is advisable to obtain the complete code, for example from its repository, and evaluate it as in the previous case.

```
# Instructions for evaluating an open educational resource (VCER evaluation)

These instructions come from the guide «Responsible vibe coding», for
publishing educational materials created with vibe coding
(https://vibe-coding-educativo.github.io/vibe-responsable/en/).
They are for evaluating an existing resource, one's own or someone else's,
whether or not it was created following the guide.

## How to evaluate

- Work on the code of the resource I give you. If you do not have it, ask me
  for it before starting.
- Do not fix it: only evaluate it.
- Score each of the ten points with 2, 1 or 0 according to the VCER rubric
  (responsible educational vibe coding, from its Spanish initials), and
  justify each score with a sentence about what you have seen in the
  resource.
- If a point cannot be checked with what I have given you, leave it
  unscored and say what would be needed to check it.
- Before scoring, make a complete inventory, not a sample: every function
  that asks for or stores data about people and how it is exported or
  copied; every image, sound, video, icon and typeface in the project, and
  where it comes from (look at the images: a redrawn logo is still someone
  else's); and everything that is loaded from outside, and when (on opening,
  when something is pressed or in a new window).
- If the resource is published, evaluate that version or check that it
  matches the code you have been given. Record which version you have
  evaluated: the version tag and the commit if it is in a repository, or the
  address and the date of the evaluation if it is not.
- For point 5, if you can run code, open the resource in a browser and run
  an automated accessibility tool on it, such as axe-core, also with content
  loaded and not only on the initial screen. Do not count what is embedded
  from other sites. The tool does not check keyboard use: go through it with
  the Tab key. In the justification of the point, say what you have tested
  and how; if you have only been able to read the code, say so.
```

```
## VCER rubric

1. CONTENT (disqualifying)
   2: No errors are detected in what it teaches: the data, the definitions
      and the answers it accepts as correct.
   1: There is some minor inaccuracy in what it teaches.
   0: There are obvious errors in concepts, data or answers.
   This score only reflects the errors you have detected: list each one and
   point out what a person should check. Typos and formatting errors in the
   references do not count here: point them out separately.

2. PERSONAL DATA (disqualifying)
   2: It does not ask for data that identifies anyone, or it stores them
      only on the device and allows exporting without names; or it sends
      them only to a school service, with no keys visible in the code and in
      such a way that only the teacher or the school can read them. It has
      no analytics; at most, an aggregated count of visits per page, kept by
      the publisher on their own server and explained on the website,
      storing nothing that tells readers apart, neither their IP address nor
      where they came from.
   1: It does not send students' data to outside services, but it stores
      names without an option to export without them, or the sending to the
      school's service is not well protected (keys in the code, an address
      that allows what is collected to be read).
   0: It sends students' data to a server outside the school, or it has
      analytics or tracking: a third-party service, or a count that tells
      readers apart or stores their IP address or where they came from.
   If the resource handles real students' data, state that a review by a
   person with technical knowledge is advisable before using it.
   What is loaded or embedded from other servers (videos, audio, typefaces,
   libraries) is assessed in point 4, not here.

3. UNDERSTANDING WHAT IT DOES
   2: What it does, what it stores and what it communicates with can be
      described briefly, and the resource's statements match its code.
   1: Its operation can be understood, but there are parts whose purpose is
      unclear or it does not explain what it stores.
   0: There are functions or communications whose purpose cannot be
      explained, or what it declares does not match the code.

4. DEPENDENCIES
   2: What it loads or embeds from outside comes from well-known services
      and is recorded, and its own texts, images and data are inside the
      material.
   1: It loads or embeds external resources without recording them, or part
      of its own content (a video, an audio recording, a map) is only on another
      platform.
   0: The main content depends on an external service, or it loads code from
      unknown addresses.

5. ACCESSIBILITY
   2: It can be used entirely with the keyboard, the controls have labels,
      the images alternative text, the contrast is sufficient, it does not
      rely on colour and it adapts to a narrow screen.
   1: It fails in one or two of those aspects, without preventing its use.
   0: It fails in three or more, or it cannot be used with the keyboard.
```

```
6. THIRD-PARTY MATERIAL
   2: Each third-party element states its author, source and licence, and
      the licence allows it to be reused; or there is no third-party
      material.
   1: There is third-party material with a valid licence, but not fully
      credited.
   0: There is third-party material without a licence that allows reuse or
      without any attribution.

7. RECORD
   2: There is a log or a note with the important decisions and their
      reasons.
   1: There is documentation explaining how it is made, but not why.
   0: There is nothing.

8. USE OF AI
   2: It states that it was made with AI and what the person has checked.
   1: It states that it was made with AI, without saying what has been
      checked.
   0: It does not state it. If it is stated that no AI was used, it is not
      scored.

9. LICENCE
   2: Authorship and a free licence visible in the material, with a link. If
      it is a multi-file project, it includes the licence file.
   1: The authorship or the licence is missing, the licence is not free (NC
      or ND) or it does not link to its text.
   0: It states neither authorship nor licence.

10. REUSE
   2: The code can be obtained in full, it is understandable when read, with
      comments where needed, and there are instructions for modifying it.
   1: It can be obtained, but it is hard to understand without comments, it
      is partly compressed or it has no instructions for modification.
   0: It cannot be obtained, or it is obfuscated or compressed.

## Result

- Calculate the percentage: the sum of the scores divided by the maximum
  possible for the points scored.
- Give a result: «Not recommended» if point 1 or 2 has a 0, whatever the
  percentage; «Needs improvement» below 70 %; «Recommended» from 70 %, as
  long as points 1 and 2 could be scored and point 5 does not have a 0. If
  either of points 1 and 2 has been left unscored, the result is «Needs
  improvement» and states what remains to be checked; if point 5 has a 0, it
  is also «Needs improvement» and states what prevents its use.
- Start the report with a line such as «VCER rubric: Recommended (85 %)»,
  followed by another with the version evaluated, and add below it a
  sentence with what that result means:
  - Recommended: it meets the essentials of the guide and can be used or
    published; the proposed improvements complete it.
  - Needs improvement: it has flaws that should be fixed before publishing
    or recommending it, although none of them rules it out.
  - Not recommended: it has obvious errors in what it teaches or sends
    students' data to services outside the school; it should not be used or
```

```
    published until it is fixed.
- In the justification of point 3, include that brief description. In that
  of point 4, list the external addresses in the code and what each one is
  for, and say in simple words what would stop working when opening the
  downloaded copy on a computer without an internet connection. In that of
  point 6, list the third-party material.
- End with the three improvements that would raise the score the most.

## If the resource is mine

- Do this only if the resource is mine and you can modify its files. If you
  do not know whether it is mine, ask me when you finish the report. If it
  belongs to someone else or you cannot modify its files, do not mention it.
- Offer me these two things and do only the ones I approve:
  - If the project has its own folder, such as a repository, save the full
    report in a file called evaluacion-vcer.md, which replaces the one from
    a previous evaluation.
  - Add to the footer of the material, in its language, a mention with the
    result, such as «VCER evaluation of version 1.2: Recommended (85 %),
    October 2026», linked to an address like this one:
    https://vibe-coding-educativo.github.io/vibe-responsable/vcer/?
r=recomendable&p=85&f=2026-
10&v=1.2&t=Equivalent%20fractions&u=https%3A%2F%2Fejemplo.github.io%2Ffracciones%2F
- In the link, r is the result (recomendable, mejorable or no-recomendable,
  always written this way); p, the percentage; f, the year and month of the
  evaluation; v, the version evaluated; t, the title of the resource, the
  same one that appears in its footer, and u, its address. Encode t and u as
  is done in a web address.
- If the resource has no version number, remove «of version 1.2» from the
  mention and v from the link. If it is not published on the web, remove u
  from the link.
- The percentage is optional: offer me the mention with it and without it.
  Without the percentage, remove «(85 %)» from the mention and p from the
  link.
- If there is already a mention, update it with the new result.

## If I then ask you to fix it

- Before changing anything, tell me what you would change and wait for me
  to approve it. Do not change the content or how it works, unless I ask
  you to.
- If there is no decision log, write one that describes how the resource
  works now, and state that it was written afterwards.
- If the licence or the statement on the use of AI is missing, prepare the
  text for me to add it.
```

The AI scores each recommendation with the VCER rubric, gives a final percentage and proposes the three improvements that would raise the score the most. If it is then asked to fix it, it first proposes the changes and waits for them to be approved; it is advisable to save a copy beforehand. If the resource is one's own, the AI also offers to save the report in the project and to add to the footer of the material a mention with the result, linked to this page.

The VCER rubric

Each point of the rubric checks one of the recommendations of the guide, which is developed in its chapter. This is the rubric used by the evaluation file:

VCER rubric: 2, met; 1, partly met; 0, not met	
1 Content (disqualifying)	
2	No errors are detected in what it teaches: the data, the definitions and the answers it accepts as correct.
1	There is some minor inaccuracy in what it teaches.
0	There are obvious errors in concepts, data or answers.
2 Personal data (disqualifying)	
2	It does not ask for data that identifies anyone, or it stores them only on the device and allows exporting without names; or it sends them only to a school service, with no keys visible in the code and in such a way that only the teacher or the school can read them. It has no analytics; at most, an aggregated count of visits per page, kept by the publisher on their own server and explained on the website, storing nothing that tells readers apart, neither their IP address nor where they came from.
1	It does not send students' data to outside services, but it stores names without an option to export without them, or the sending to the school's service is not well protected (keys in the code, an address that allows what is collected to be read).
0	It sends students' data to a server outside the school, or it has analytics or tracking: a third-party service, or a count that tells readers apart or stores their IP address or where they came from.
3 Understanding what it does	
2	What it does, what it stores and what it communicates with can be described briefly, and the resource's statements match its code.
1	Its operation can be understood, but there are parts whose purpose is unclear or it does not explain what it stores.
0	There are functions or communications whose purpose cannot be explained, or what it declares does not match the code.
4 Dependencies	
2	What it loads or embeds from outside comes from well-known services and is recorded, and its own texts, images and data are inside the material.
1	It loads or embeds external resources without recording them, or part of its own content (a video, an audio recording, a map) is only on another platform.
0	The main content depends on an external service, or it loads code from unknown addresses.
5 Accessibility	
2	It can be used entirely with the keyboard, the controls have labels, the images alternative text, the contrast is sufficient, it does not rely on colour and it adapts to a narrow screen.
1	It fails in one or two of those aspects, without preventing its use.
0	It fails in three or more, or it cannot be used with the keyboard.

6 Third-party material	
2	Each third-party element states its author, source and licence, and the licence allows it to be reused; or there is no third-party material.
1	There is third-party material with a valid licence, but not fully credited.
0	There is third-party material without a licence that allows reuse or without any attribution.
7 Record	
2	There is a log or a note with the important decisions and their reasons.
1	There is documentation explaining how it is made, but not why.
0	There is nothing.
8 Use of AI	
2	It states that it was made with AI and what the person has checked.
1	It states that it was made with AI, without saying what has been checked.
0	It does not state it. If it is stated that no AI was used, it is not scored.
9 Licence	
2	Authorship and a free licence visible in the material, with a link. If it is a multi-file project, it includes the licence file.
1	The authorship or the licence is missing, the licence is not free (NC or ND) or it does not link to its text.
0	It states neither authorship nor licence.
10 Reuse	
2	The code can be obtained in full, it is understandable when read, with comments where needed, and there are instructions for modifying it.
1	It can be obtained, but it is hard to understand without comments, it is partly compressed or it has no instructions for modification.
0	It cannot be obtained, or it is obfuscated or compressed.

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Credits and licences

Publication and licences

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How it was made

This guide was produced at level 4 of the Framework for integrating generative AI into educational tasks (MIAE) (https://jjdeharo.github.io/miae/?nivel=4), which corresponds to advanced collaboration between the person and the AI. The text was written in dialogue with Claude Code, and the author directed and corrected the content, checked the claims against their sources and reviewed the result. The translations into Catalan, Galician, Basque and English were made automatically with the same AI, from the Spanish original, and have not been reviewed by a professional translator.

How to cite

De Haro, J. J. (2026). *Responsible vibe coding: a guide to publishing educational materials created with vibe coding* (version 3.0). Vibe Coding Educativo. https://doi.org/10.5281/zenodo.24073199 (https://doi.org/10.5281/zenodo.24073199)

To cite the guide without specifying a particular version: https://doi.org/10.5281/zenodo.23081517 (https://doi.org/10.5281/zenodo.23081517), which always leads to the latest one.

Versions

The guide is numbered with two figures. The first changes when what the guide says changes, and the second when the wording or the translations are corrected without changing the content. Each version has its own DOI.

- **3.0**, 9 November 2026. DOI: 10.5281/zenodo.24073199 (https://doi.org/10.5281/zenodo.24073199).
    - Data protection allows at most a count of visits per page that does not tell readers apart, kept by the publisher on their own server and explained on the website, like the one the AEPD exempts from requiring consent.
    - The VCER rubric (point 2), chapter 2 and the instructions for creating with AI cover this case; any other analytics or tracking still scores 0.

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  - The VCER evaluation has its own page, with the evaluation file, the rubric and the explanation of the result.
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  - Corrects the translations and the wording without changing what the guide says.
  - States in the credits that the translations are automatic and have not been reviewed by a professional translator.
- **1.0**, 1 October 2026. DOI: 10.5281/zenodo.23081518 (https://doi.org/10.5281/zenodo.23081518).
  - First published version.

Third-party materials

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The documents consulted to prepare the guide, with their authorship and licence, are listed in the index of sources (https://github.com/Vibe-Coding-Educativo/vibe-responsable/blob/main/fuentes/README.md) in the repository.

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Ernesto Serrano, from the eXeLearning team, proposed three improvements that the guide has incorporated: turning the review into checks (https://github.com/Vibe-Coding-Educativo/vibe-responsable/issues/2) that are repeated after each change (recommendation 1), completing the decision log (https://github.com/Vibe-Coding-Educativo/vibe-responsable/issues/1) with the evidence, risks and validation of each technical decision (recommendation 7) and identifying each published version (https://github.com/Vibe-Coding-Educativo/vibe-responsable/issues/3) with a tag (recommendation 10).

David Cordones pointed out in the Vibe Coding Educativo group (https://t.me/vceduca/13553/13587) that the guide could be overwhelming for beginners, which led to simplifying the introduction, and asked about the NC condition of the licences, which chapter 9 now explains in more detail.